Intelligent Verification System for Concrete Compliance & QA Risk Management

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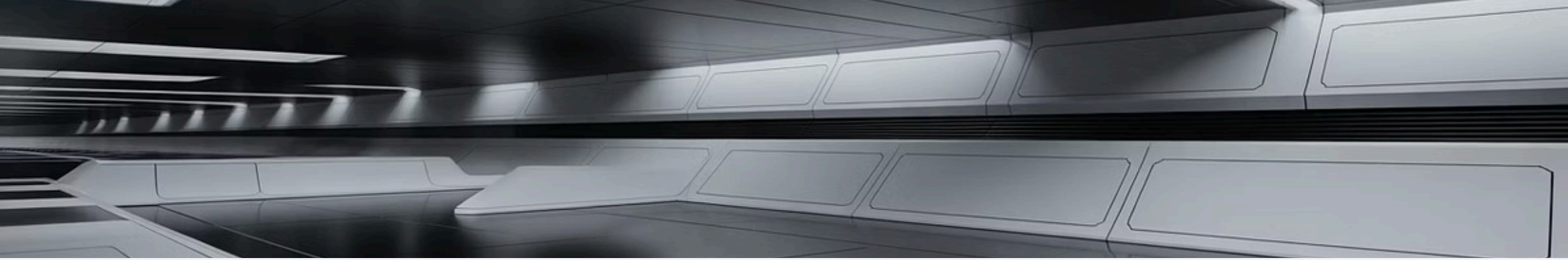
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Section 1: Company Overview

Company Overview

Ambient Systems is an AI-driven compliance and sustainability venture re-establishing Singapore operations to pilot BEAMP collaborations. We combine cutting-edge artificial intelligence with deep domain expertise to transform how the built environment manages quality assurance and regulatory compliance.

Spun-off from the National University of Singapore in 2021, we've rapidly expanded our with operations spanning **Serbia (HQ), Hong Kong**, and Quebec. Our technology has been validated across multiple international markets and recognized by leading innovation platforms.

Recognition & Innovation

- **WIPO Patent WO** — Proprietary AI compliance framework
- **Panasonic Deep Tech Challenge Winner** (2023)
- **Google Runner-up** (2022)



Business Activities

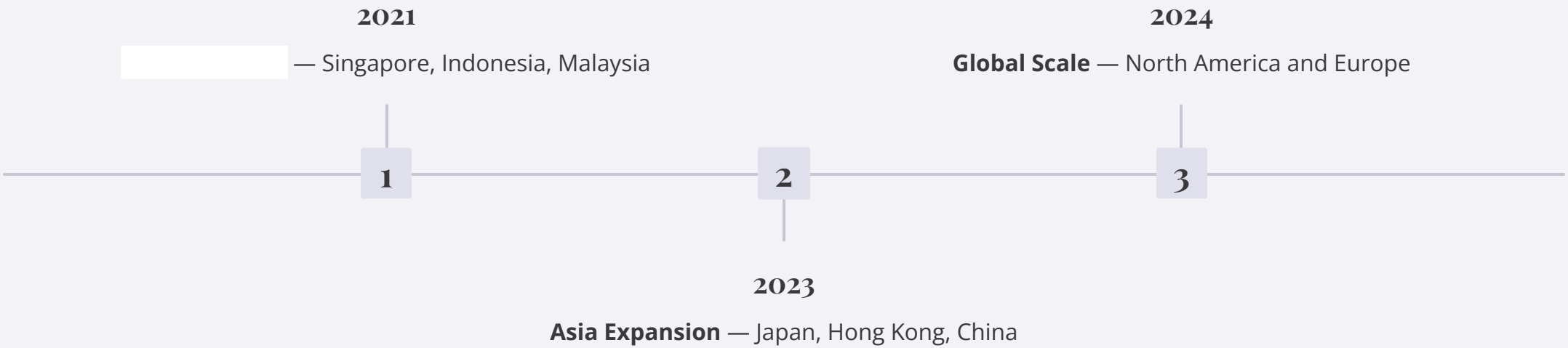
Ambient Systems develops **AI systems for compliance automation, ESG intelligence, and performance verification**. Our technology serves as the intelligent backbone for modern construction quality assurance, transforming fragmented data into actionable insights.

Core Verticals

- Construction QA automation
- Infrastructure digital twins
- Sustainability finance analytics

- environment localization
- FinTech/GreenTech/ConTech integration
- Automated verification systems

Global Market Presence



Singapore serves as our strategic testing and validation hub, enabling us to build robust solutions before scaling globally.



Products / Services

We offer three integrated offerings specifically designed to address the challenges outlined in the BEAMP Cycle 6 program. Each solution builds upon our core AI and compliance technology platform.

1. POC / Pilot — BEAMP Cycle 6 Focus

Our pilot program delivers targeted AI-driven verification solutions addressing critical pain points in construction quality assurance and compliance management.

Concrete Quality

Automated verification of delivery orders, slump tests, and cube-test data. Our system ingests data from multiple sources — WhatsApp logs, PDFs, spreadsheets — and reconciles them into a unified compliance record.

Construction Economics

Intelligent linking of QA performance to financial metrics and project milestones. Track how quality compliance impacts cost, schedule, and payment approval workflows in real-time.

Sustainability Metrics

Comprehensive tracking of embodied carbon and materials data for ESG reporting. Transform QA data into sustainability insights that meet Singapore's Green Building standards and international ESG frameworks.



2. Platform (ambientsystems.ai)

Our comprehensive platform integrates quality assurance, financial performance, and sustainability analytics into a unified intelligence system. Built on cutting-edge AI and designed for seamless integration with existing construction technology stacks.

Financial Metrics

Seamlessly integrates QA and project performance data with cost visibility dashboards. Track how quality compliance impacts your bottom line.

scenario modeling for compliance forecasting and construction logistics optimization. Predict potential issues before they occur.

IoT Connectivity

Direct interfaces with sensors and delivery data systems enable automated real-time verification without manual data entry.

The platform serves as the central nervous system for construction compliance, connecting disparate data sources and transforming them into actionable intelligence for project teams, auditors, and stakeholders.

3. Consulting (ivandamnjanovic.com)

Beyond technology, we provide deep technical consulting services that leverage our founder's extensive expertise across multiple engineering domains and intellectual property development.



Civil Engineering

Structural QA, performance audits, and digital twin modeling for infrastructure projects. We bring rigorous engineering analysis to compliance verification.



Energy Systems

Renewable energy integration, load optimization, and carbon analytics. Helping projects achieve net-zero targets through data-driven strategies.



Controlled Environment Agriculture

Data-driven optimization of indoor production systems, applying our AI expertise to sustainable food production.



Intellectual Property

Systems design and patented AI compliance frameworks (WO). We help organizations protect and monetize their innovations.

Business Targets

1-3 Year

Intellectual Property Establishment

Secure Singapore-based IP portfolio and regulatory approvals

Complete Platform Development

Finalize core modules and achieve production-ready status

Client Tester Acquisition

Secure LOIs/MOUs and advance Technology Readiness Level from 6 to 9

Singapore Entity Strategy

Ambient Systems is establishing a dedicated Singapore business focused on **applied AI for compliance, sustainability, and performance verification** in the built environment. This new entity will leverage our existing R&D and IP portfolio (including patent WO) while serving as a Singapore-based venture specifically designed to co-develop and pilot solutions with BEAMP ecosystem partners.

Global Validation

Our prior operations across **Canada, Hong Kong, and Serbia** have validated the urgent market need for integrated ESG-AI systems. The **BEAMP programme** offers the ideal opportunity to anchor these proven capabilities in Singapore, then scale solutions to clients across the globe.

Key Team Members

Ivan Damnjanovic

Founder & Systems Architect

Entrepreneur / Inventor based in Belgrade and Hong Kong with 5+ years across Asia (Singapore, Nepal, Hong Kong). Brings **10 years of experience** in technical management, risk analysis, and AI deployment.

- R&D expert with projects under NUS, NSERC, and Centennial College
- PhD-level expertise in Business Management & Economics
- Track record scaling YC-backed startups and cross-border ventures
- Patent holder (WO) and lead architect of Ambient Verify and Ambient Twin

Languages: Native Serbian & English (fluent); Intermediate French

David Damnjanovic

Advertisement Specialist

Advertising specialist at **Yourbow.com**, bringing commercial marketing expertise and go-to-market strategy to the venture. Based in Belgrade, Serbia.



Section 2: Proposal

Proposal Summary

Challenge Statement (CS 9): Intelligent Verification System for Concrete Compliance and QA Risk Management

Challenge Statement Owner (CSO): Expand Construction Pte. Ltd.

Our Solution

Ambient Systems proposes an **AI-driven verification platform** that automates data ingestion, reconciliation, and compliance verification across diverse document formats — PDFs, WhatsApp logs, spreadsheets, and e-delivery orders. The system unifies QA/QC, finance, and sustainability datasets into a single dashboard for real-time traceability and ESG-linked performance assurance.

Technology Core

Unified AI engine combining **OCR, NLP, and machine learning** for automated QA verification. Reconciliation logic linking Delivery → Slump → Cube → Approval workflows. Role-based dashboards delivering instant anomaly detection and ESG metrics.

Key Deliverables

- Real-time traceability & audit readiness
- Structured QA datasets for compliance & ESG analytics
- Intelligent risk alerts for claims and approvals

End-to-End Coverage

From delivery orders through testing to audit readiness, with financial and compliance considerations integrated at every stage of the process.

Introduction

The Problem Context

Concrete QA data today is **fragmented across WhatsApp, Excel, eDO portals, and emails** — making reconciliation slow, error-prone, and unverifiable at audit time. Expand Construction currently manages **100–200 delivery orders per pour**, each requiring manual cross-checks between DOs, cube tests, and invoices. This creates significant operational inefficiency and compliance risk.

Solution Functions & Features

1

AI/NLP Engine

Extract key fields from semi-structured documents automatically

2

Reconciliation Logic

Link DO → Slump → Cube → Approval with intelligent matching

3

Role-Based Dashboards

Tailored interfaces for QA, Finance, and Audit teams

4

Configurable Alerts

Proactive detection of missing or non-compliant records

5

Seamless Integration

Works with Autodesk Construction Cloud & supplier portals

Value Proposition

- Reduces QA collation time by up to **70%**
- Ensures **100% traceability** across workflows
- Supports Singapore's Construction 4.0 and ESG standards

Competitive Advantages

- Engineering expertise + explainable AI
- Patented IP (WO)
- Ambient Verify + Ambient Twin integration

Integration Philosophy

Designed to **enhance existing tools** — not replace them. Works alongside your current technology stack.

Technology Architecture

Our solution is built on a sophisticated multi-layer architecture that combines cutting-edge AI with robust engineering principles. Each component is designed for scalability, security, and seamless integration with existing construction technology ecosystems.

AI / NLP Ingestion Engine

Automated parsing of WhatsApp, PDF, Excel, and e-DO data; extracts key fields (DO, slump, cube ID, strength)

Tech Stack: OCR + NLP (LayoutLM / GPT) · Python · Pandas

Reconciliation Logic

Links DO → Slump → Cube → Approval; detects mismatches and anomalies with intelligent pattern recognition

Tech Stack: SQL + Rule-Based Engine + ML Consistency Model

Alert & Workflow System

Real-time detection of missing data or non-compliance; triggers notifications to relevant stakeholders automatically

Tech Stack: Event Triggers · Threshold Rules · Automation Scripts

Role-Based Dashboards

Interactive visualizations for QA, Finance, and Audit teams; exportable audit reports with full traceability

Tech Stack: React / Streamlit · Supabase / PostgreSQL

ESG Analytics Layer

CO_2 and energy traceability; sustainability & carbon-linked KPIs powered by Ambient Twin

Tech Stack: Ambient Twin API · Carbon Factor Database · ESG Models

Security & Compliance

Access control, encryption, data audit logs; CSA-aligned data protection standards

Tech Stack: OAuth 2.0 · AES-256 · CSA Cyber Essentials (SG GovTech)

POC / Pilot Plan

Pilot Overview

- **Objective:** Pilot AI-based verification on active project (NS Square)
- **Duration:** 4 months core development + 2 months validation
- **Budget:** SGD 40,000 – 50,000
- **Team:** 2 Data Scientists · 1 Systems Engineer · 1 QA Specialist

Stakeholders & Roles

Ambient Systems: AI system design, implementation, and technical support

Expand Construction: Data access, site validation, feedback loop, and domain expertise

BEAMP Ecosystem: Mentorship, funding support, and network connections

Phase 1 — Month 1

1

Data mapping & sample collection (WhatsApp, PDFs, Excel). Establish data pipelines and validation protocols.

Phase 2 — Month 2

2

AI ingestion prototype development + automated data reconciliation engine build-out.

Phase 3 — Months 3-4

3

Dashboard deployment & live testing with real project data from NS Square.

Phase 4 — Month 5

4

Validation & ROI assessment. Measure time savings and accuracy improvements.

Phase 5 — Month 6

5

Integration into Expand's ecosystem. Handover and training for full adoption.

Impact

Quantitative Benefits

70%

Time Reduction

Reduction in manual QA collation time — from days to hours

2

Days Saved

Working days saved per month per QA team through automation

100%

Traceability

Complete traceability for audit compliance and claims management

Qualitative Impact

ESG Excellence

Real-time ESG and CO_2 tracking via Ambient Twin layer enables comprehensive sustainability reporting aligned with Singapore's Green Building standards.

Financial Confidence

Greater confidence in milestone payments and audit readiness reduces financial risk and accelerates project cash flow.

Staff Empowerment

Empowers staff with "AI personal assistants" that handle repetitive tasks, allowing them to focus on high-value strategic work.

Manual Process

- Fragmented data across multiple platforms
- Hours of manual reconciliation per pour
- High error rates and missing records
- Delayed audit preparation

Intelligent QA Verification

- Unified data dashboard with real-time updates
- Automated reconciliation in minutes
- 99%+ accuracy with anomaly detection
- Audit-ready reports on demand

Scalability

Our vision extends far beyond the initial pilot. We're building a scalable platform architecture that can expand across geographies, sectors, and compliance domains to become the industry standard for intelligent verification.

Singapore Entity

New Ambient Systems Singapore venture focused on AI compliance tools for the built environment. Anchor operations and establish regional hub.

Regional Scaling

Deploy in Hong Kong and Canada for cross-jurisdiction testing and commercial proof. Validate technology across diverse regulatory environments.

Sector Expansion

Extend to steel verification, prefabrication QA, and BIM compliance modules. Create comprehensive construction verification suite.

Long-Term Vision

Create an **interoperable digital twin for compliance** — a unified QA and ESG verification layer for Construction 4.0. This will serve as the intelligent infrastructure enabling the next generation of smart, sustainable, and compliant building practices globally.

Singapore Hub

R&D and BEAMP pilot

Hong Kong

Asia-Pacific expansion

Europe

EU compliance standards

Canada

North America deployment

Turning Fragmented QA Data into Real-Time Compliance Intelligence

Re-establishing **Ambient Systems (Singapore)** to lead the next wave of **AI + ESG transformation** in the built environment.

Empowering contractors, auditors, and regulators with **data-driven verification and sustainability insights**.



Contact

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"From delivery orders to ESG dashboards — making compliance measurable, verifiable, and intelligent."